

Isaac Herbst

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LEAD SOFTWARE ENGINEER | CLOUD, AI & DISTRIBUTED SYSTEMS

SUMMARY

Hands-on technical lead with experience architecting healthcare software platforms, AI-powered retrieval systems, and scalable AWS infrastructure. Designed and built production RAG pipelines, distributed processing systems, and operational tooling using AWS serverless technologies, OpenSearch vector databases, and modern full-stack frameworks. Helped scale engineering operations from 2 to 8 team members in a fast-moving startup environment.

TECHNICAL SKILLS

AI & Infrastructure: RAG Architecture, Vector Search, Embeddings, AWS Lambda, EC2, S3, SQS, DynamoDB, OpenSearch

Backend: Node.js, Python, C#, REST APIs, Distributed Systems, Serverless Architecture

Frontend: React.js, JS.TS, MUI, HTML/CSS

Databases & Tools: SQL Server, MySQL, Docker, Git, Playwright, CloudWatch, CodeDeploy

AI-Assisted Development: Claude Code, skills, agents, MCP

EXPERIENCE

Lead Software Engineer – PreciseMDS | 2024 – Present

- Lead architecture and development of healthcare data platforms, AI-powered retrieval systems, and AWS cloud infrastructure.
- Designed and implemented production RAG systems using vector retrieval pipelines and embedding workflows.
- Architected distributed AWS infrastructure using Lambda, SQS, EC2, S3, DynamoDB and more to support scalable concurrent processing.
- Architected the frontend and developed a Chrome extension integrating with PCC.
- Built high-throughput document ingestion and AI-assisted automation systems for healthcare operational workflows.
- Helped scale the engineering organization from 2 to 8 team members while contributing to technical direction and mentoring.
- Drive infrastructure reliability, deployment workflows, monitoring, and production troubleshooting across distributed systems.

Software Developer – CAB (Central Analysis Bureau) by Randall Reilly | 2022 – 2024

- Developed and maintained full-stack applications focused on data analysis and visualization.
- Modernized legacy systems with React-based user interfaces and improved usability.
- Worked across frontend, backend, and database layers using React, JavaScript, ColdFusion, SQL Server, and AWS.

Software Developer – Lakewood Matzah Bakery | 2021 – 2022

- Built internal reporting and operational systems and automated business workflows.

Mentor – Krovei Lev | 2019 – Present

- Mentor children through community support and educational initiatives.

EDUCATION

PCS – Software Development Course | Class President

Comprehensive software engineering program focused on modern full-stack development.

Udemy – Cryptography & Blockchain Full Stack Development

Built blockchain applications using Node.js, Express.js, React.js, and Jest.

Beth Medrash Govoah – Master's in Talmudic Law

Developed advanced analytical reasoning and problem-solving skills.